The Hong Kong Polytechnic University
Department of Building Environment and Energy Engineering

BSE3124 Electricity and Buildings **2026/27 Semester 1**

Assignment – Answer Sheets

Group number: _____

Names of group members: _____

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Task 1 Loading estimation and transformer room location

Working submission v01. Official cover and task structure are retained. The stated assumptions define a course design, not site certification.

Building load assessment

Adopt the drawing areas 309, 309 and 410 m^2 as usable floor areas for ten identical office floors. This is a course assumption: the UFA exclusions have not been independently reconciled. Use supplied CLP COP215 Rev 07. The assignment's 1000 kW is cooling capacity. Assume packaged air-cooled central AC, electrical PF 0.90 and five lifts. The 40 kVA/lift public-services allowance is aggregate, not a lift motor rating.

$$
\begin{aligned}
A_{\text{UFA}} &= 10(309 + 309 + 410) = 10\,280\ \mathrm{m}^2, \\
S_{\text{office}} &= 10\,280(0.16) = 1644.800\ \text{kVA}, \\
P_{\text{AC}} &= \frac{1000}{3.51685}(1.85) = 526.039\ \text{kW}, \\
S_{\text{AC}} &= 526.039/0.90 = 584.488\ \text{kVA}, \\
S_{\text{public}} &= 5(40) = 200.000\ \text{kVA}.
\end{aligned}
$$

COP215 Appendix 2 supplies the office density; Appendix 7 supplies the AC reference. Public services follow clauses 3.1.6 and 3.3.1 and include lobby AC. Let X kVA be the duplicated AC share. Because it cannot exceed either overlapping allowance, $0 \leq X \leq 200$. For the proposed two independent supply zones, apply clause 3.1.7:

$$
S_{\text{assessment}} = 1.1(1644.800 + 584.488 + 200 - X).
$$

The range is 2452.216–2672.216 kVA. Adopt the upper bound **2672.216 kVA** for capacity reservation. This does not assert that actual overlap is zero. Do not add public lighting/pumps/fire-service allowances again on top of the aggregate reference.

Transformer and main LV room proposal

Propose two 1500 kVA transformers with separate LV sections:3000 kVA total gives 327.784 kVA margin. Allocate floors 1–5 to A and 6–10 to B, each with half the central plant/public allowance. Each section is 1336.108 kVA, about 2030 A at 380 V. The equal split is assumed. Full-load N–1 is not provided; load shedding is required on loss of one unit. Any coupler must prevent parallel operation. Place the rooms at the lower-left perimeter of the supplied plan: substation 12430 by 5450 mm and adjacent LV room 12430 by 3500 mm, the latter depth assumed. Reasons are external plant access, short LV interfaces, separation from the lift/stair/AHU core and independent LV access. Reserve a 3 m delivery route within 20 m of unloading, subject to external site confirmation. The appendix shows assumed HV3000 by 1000 mm and transformer 2200 by 1800 mm equipment reservations and six LV cabinets; these are not manufacturer footprints. Their clearance envelopes fit without overlap. COP101 v 16 clause 4.3 requires HV rear 1000/front 1800/sides 750 mm and transformer LV terminals 900/others 750 mm. Propose 2-hour partitions, sealed penetrations and FFL150 mm above external level. Outward 180-degree doors have shared sweeps requiring sequential use. Two 600 by 2000 mm louvres at assumed 50% free area provide 1.20 m^2, above clause 6.3's 1.12 m^2 reference; certified free area must be at least 46.67%. Fan symbols are conditional provisions, not actual heat-loss calculations. LV interfaces including assumed tails are 10.03/10.31 m, below the 20 m limit in 5.5.2.2. Actual equipment, structures, openings and ventilation must satisfy these declared conditions.

Task 2 Layout drawings and MCB board schematics

The office tenant chosen for layout design: Office 1.

Normal lighting and local controls

The design contains 92 working 600 mm-grid panel positions using PIL PLATO 600 MultiPower 3117955: sixty 22 W and thirty-two 28 W, total 2216 W. The retained LDT manifest gives nominal flux 2843.45 and 3555.30 lm respectively. Match the ceiling grid, partitions and furniture, with separate enclosed-space circuits and distributed main-office circuits. A500 lx task-plane target is a design assumption; certified photometric uniformity/glare compliance is not claimed. Connected lighting density at the provisional 309 m^2 is 7.172 W/m^2; this arithmetic alone is not energy-code compliance. Seven local passive manual switches S-A–S-G control their corresponding circuits. Specify minimum 20AX, LED switching at least 600 W and make/break/inrush compatibility for the selected drivers. S-A/B/C use separate enclosures because their phases differ. No borrowed neutral is allowed. Routes visit each switch; include 2 m DB rise plus 1.5 m down/1.5 m back at the switch, giving 5 m assumed vertical allowance per lighting radial. Passive controls add no operating power.

Circuit	Lamps	W	Phase	RCBO	Control
LC-A	18	402	L1	B10	S-A
LC-B	17	392	L2	B10	S-B
LC-C	18	396	L3	B10	S-C
LC-D	11	242	L3	B10	S-D
LC-E	14	392	L2	B10	S-E
LC-F	10	280	L1	B10	S-F
LC-G	4	112	L3	B10	S-G

Plan A/B/C numeric labels map to corresponding LED IDs and LC-A/B/C. D01–D11 map to PN01–PN11, E82–E96 excluding E90 to northeast LEDs, F01–F10 to UE01–UE10, G01–G04 to SE01–SE04. Power settings follow the common device schedule. Every luminaire and control has one circuit assignment.

Small power and fixed equipment

Provide nine four-socket floor boxes, three twin wall sockets and fixed connection FS01. Each floor-box/socket assembly is individually 13 A fused. This defines intended continuous demand, not an instantaneous current clamp. FS01 reserves equipment up to 13 A; appliance identity/watts are not invented. Building central AC is supplied as central plant and not duplicated in this office DB. P1–P4 are A1 rings with fused floor-box connections. P5/P6/P8 are A2 radials; P7 is a fixed radial. Both P6 and P8 serve the source PANTRY; including FB09 and SO01. P8 is retained as a separate socket group by design choice, not because each pantry outlet requires its own final circuit. Kitchen sockets are separated from ordinary office circuits under EMSD6E(3)(a).

Circuit	Points	Phase	Topology	Protection
P1	FB01, FB02	L1	A1 ring	B32
P2	FB03, FB04	L1	A1 ring	B32
P3	FB05, FB06	L2	A1 ring	B32
P4	FB07, FB08	L2	A1 ring	B32
P5	SO02	L1	A2 radial	B32
P6	FB09, SO01	L3	A2 radial	B32
P7	FS01	L3	Fixed radial	B16
P8	SO03	L3	A2 radial	B32

Defined service envelopes satisfy A1 maximum 100 m^2 and A2 maximum 50 m^2. These are designed coverage polygons, not net UFA, with intentional overlap. Ring L/N/CPC return to their own circuit terminals. Main 6/10 mm^2 cables terminate in suitable accessible junction/FCU assemblies, not assumed ordinary socket terminals. Local connections use 4 mm^2 L/N/CPC and assumed 0.5 m per point.

Protection and board arrangement

The complete schematic shows C63 four-pole protection at MS-O1, 4-core 35 mm^2 plus CPC16, a 100 A four-pole DB isolator, independent L1/L2/L3/N/PE bars and 15 separate 1P+N TypeA30mA RCBOs. Every load neutral passes through its own RCBO; none are shared downstream. PE bypasses all switches. Label circuits/phases and allow one final circuit to be isolated without interrupting the rest. Use B-curve final protection and require compatible driver inrush. Do not change to a C curve without recalculation. Apparatus must meet Task 3 breaking/energy limits. Ratings/curves alone do not prove full short-circuit discrimination; none is claimed.

Emergency lighting assumption

Assume landlord emergency lighting/exit signage on independent essential systems, electrically separated under EMSD6B(4). No ordinary tenant panel is an emergency fitting. Emergency feeds must not pass through tenant normal-lighting switches or these RCBO load sides. Emergency load is excluded from this normal DB under this explicit course boundary; installed escape coverage, autonomy and illuminance are not certified.

Task 3 Maximum current demand calculation and cable sizing

Maximum current demand

For lighting, $I_b = P/(220 \times 0.90)$. Incoming diversity follows EMSD Table 7(1):90% of office lighting, and 100% of the largest Code 6E circuit plus 50% of the remaining circuit currents on each phase. Reserve P7 at 100% separately. Use designed fused-point demand bounds, not breaker nameplates as appliance loads.

Phase	Lighting A	Socket demand A	P7 A	Total demand A
L1	3.444	45.500	0.000	48.600
L2	3.960	39.000	0.000	42.564
L3	3.788	32.500	13.000	48.909

Maximum phase demand is **48.909 A**; select C63 at the source and 100 A DB isolation/busbars. Maximum undiversified phase point allowance 68.444 A is below the board rating. This demand depends on the specified 13 A fused-point restrictions.

Incoming cable

Select 4-core 35 mm^2 PVC copper L1/L2/L3/N plus separate 16 mm^2 PVC copper CPC, MethodB in protected enclosed containment. Proposed same-floor meter-to-DB route 28.5119 m plus assumed 3 m total vertical/termination allowance gives 31.5119 m. The proposed source is near the supplied METER label, not a measured termination. Upstream ground-to-floor distribution is retained through explicit voltage/fault conditions. For 40 degrees C take $C_a = 0.87$, $C_g = 1$, and $C_h = 0.86$ for assumed third-harmonic content greater than 15% and at most 33% in this four-core cable (A5(7)). Keep a full-size neutral and reselect if this range is exceeded.

$$
\begin{aligned}
I_{z,25} &= 80(0.87)(0.86) = 59.856\,\text{A} < 63\,\text{A}, \quad \text{reject,} \\
I_{z,35} &= 99(0.87)(0.86) = 74.0718\,\text{A}, \\
48.9091 &\leq 63 \leq 74.0718\,\text{A}, \quad \text{pass.}
\end{aligned}
$$

TableA6(2) column 5 supplies ratings. Table 11(2)(a) gives CPC16 for $16 < S \leq 35$; energy is checked below. For unbalanced phase-neutral drop with neutral current at most 63 A, derive $(2/\sqrt{3})(1.10) = 1.270170592\,\text{mV/A/m}$ from the three-phase table impedance. Feeder drop at 63 A is 2.521606 V; at design current 1.957610 V. This fundamental-frequency check is not harmonic-distortion assessment.

All outgoing cables

All sizes are PVC copper L/N/CPC. Lighting grouped separately has $I_z = 24(0.87)(0.54) = 11.2752\text{A}$ for seven 2.5 mm^2 circuits; all $I_b \leq 10 \leq I_z$. Lighting lengths include 5 m DB/switch vertical allowance. Uniform ring banks count four current-carrying legs: $I_z = 41(0.87)(0.65) = 23.1855\text{A}$ per 6 mm^2 leg, above the standard 20 A minimum. Ring-network solution at 13 A per point gives maximum leg currents 14.658–15.203 A, below I_z. Do not compare radial $I_n \leq I_z$ against just one ring leg. Continuity and fused-point restrictions are essential. The 10 mm^2 A2 radials have $I_z = 57(0.87)(0.70) = 34.713\text{A}$; P7's separate 4 mm^2 radial has $I_z = 32(0.87) = 27.84\text{A}$. Power radial allowance is 2 m vertical, rings 2 m at each DB end; all local point spurs add 0.5 m of 4 mm^2 cable.

Circuit	Phase	I_b A	RCBO A	L/N/CPC mm^2	Main length m
LC-A	L1	2.030	10	2.5	58.301
LC-B	L2	1.980	10	2.5	62.184
LC-C	L3	2.000	10	2.5	44.999
LC-D	L3	1.222	10	2.5	27.233
LC-E	L2	1.980	10	2.5	34.464
LC-F	L1	1.414	10	2.5	32.710
LC-G	L3	0.566	10	2.5	27.379
P1	L1	26.000	32	6	28.620
P2	L1	26.000	32	6	35.520
P3	L2	26.000	32	6	45.220
P4	L2	26.000	32	6	34.220
P5	L1	13.000	32	10	15.607
P6	L3	26.000	32	10	13.864
P7	L3	13.000	16	4	13.107
P8	L3	13.000	32	10	15.107

Complete voltage drop and fault checks

Sum segment downstream-current drops on radials and solve the two-feed ring network. Include stated vertical/local allowances. Add 2.2 V pre-origin allowance,2.521606 V feeder bound and outgoing drop. The full limit is $0.04(220) = 8.8$V. Adopt TN-S, external source loop impedance 0.20 ohm and PFC at most 6 kA at MS-O1/downstream. These are course conditions, not measurements. Incoming phase+CPC bound 0.064129 ohm gives DB external loop 0.264129 ohm. Table 11(10) reference limits are B10=4.17, B16=2.61, B32=1.31 and C63=0.33 ohm. Use operating-temperature paths and minimum voltage factor 0.95. Worst uniform-ring parallel path is at most quarter circumference.

Circuit	Out drop V	Total drop %	Z_s ohm	Min fault A	Z_s limit ohm
LC-A	1.450	2.805	1.314	159.1	4.17
LC-B	1.237	2.709	1.383	151.1	4.17
LC-C	0.959	2.582	1.074	194.6	4.17
LC-D	0.422	2.338	0.754	277.1	4.17
LC-E	0.797	2.509	0.884	236.3	4.17
LC-F	0.642	2.438	0.853	245.0	4.17
LC-G	0.254	2.262	0.757	276.1	4.17
P1	1.269	2.723	0.322	649.3	1.31
P2	1.675	2.907	0.334	624.9	1.31
P3	1.937	3.027	0.352	593.5	1.31
P4	1.428	2.795	0.332	629.4	1.31
P5	0.964	2.584	0.338	617.8	1.31
P6	1.310	2.742	0.331	632.1	1.31
P7	1.946	3.031	0.414	505.1	2.61
P8	0.936	2.571	0.336	621.8	1.31

All 15 full voltage bounds pass; worst 3.031%. Each B-curve final circuit exceeds minimum earth-fault threshold $5I_n$. Incoming C63 gives $Z_s = 0.264129 < 0.33$ohm and $I_{f,\text{min}} = 791.28 > 10(63) = 630$A, satisfying reference instantaneous disconnection under the adopted conditions.

Overload and total clearing energy

Use $k = 115$ for bunched PVC copper CPCs initially 70 degrees C (11(2)(c)); require total $I^2t \leq (kS)^2$. Specify 10 kA main/RCBO interruption. Require class 3-equivalent B-curve total energy at most 70,000 A^2s for B10/B16,90,000 for B32. These are functional selection requirements, not automatic certification of any RCBO. CPC withstands are 82,656.25 (2.5 mm^2),211,600 (4),476,100 (6),1,322,500 (10), each above the required device bound. Incoming C63 total energy must be at most 1,000,000 A^2s over the relevant fault range, below CPC16 withstand 3,385,600. Complete 13 A FCU energy must be at most 90,000, below the 4 mm^2 spur withstand 211,600. The incoming/FCU limits are explicit apparatus performance specifications, not invented manufacturer data. For fuse overload, $I_2 = 1.9(13) = 24.7$A. Spur $I_z = 32(0.87)(0.70) = 19.488$A gives $1.45I_z = 28.2576 > 24.7$A. Ring leg conventional-current bounds also remain below $1.45I_z$ for the declared fused-point design. The fuse reference 6 kA interruption meets assumed PFC at most 6 kA. Its typical pre-arcing 870 A^2s is *not* total clearing energy. Assume FCU ambient at most 35 degrees C and board apparatus at most 30 degrees C; containment 40 degrees C remains the conservative cable condition.

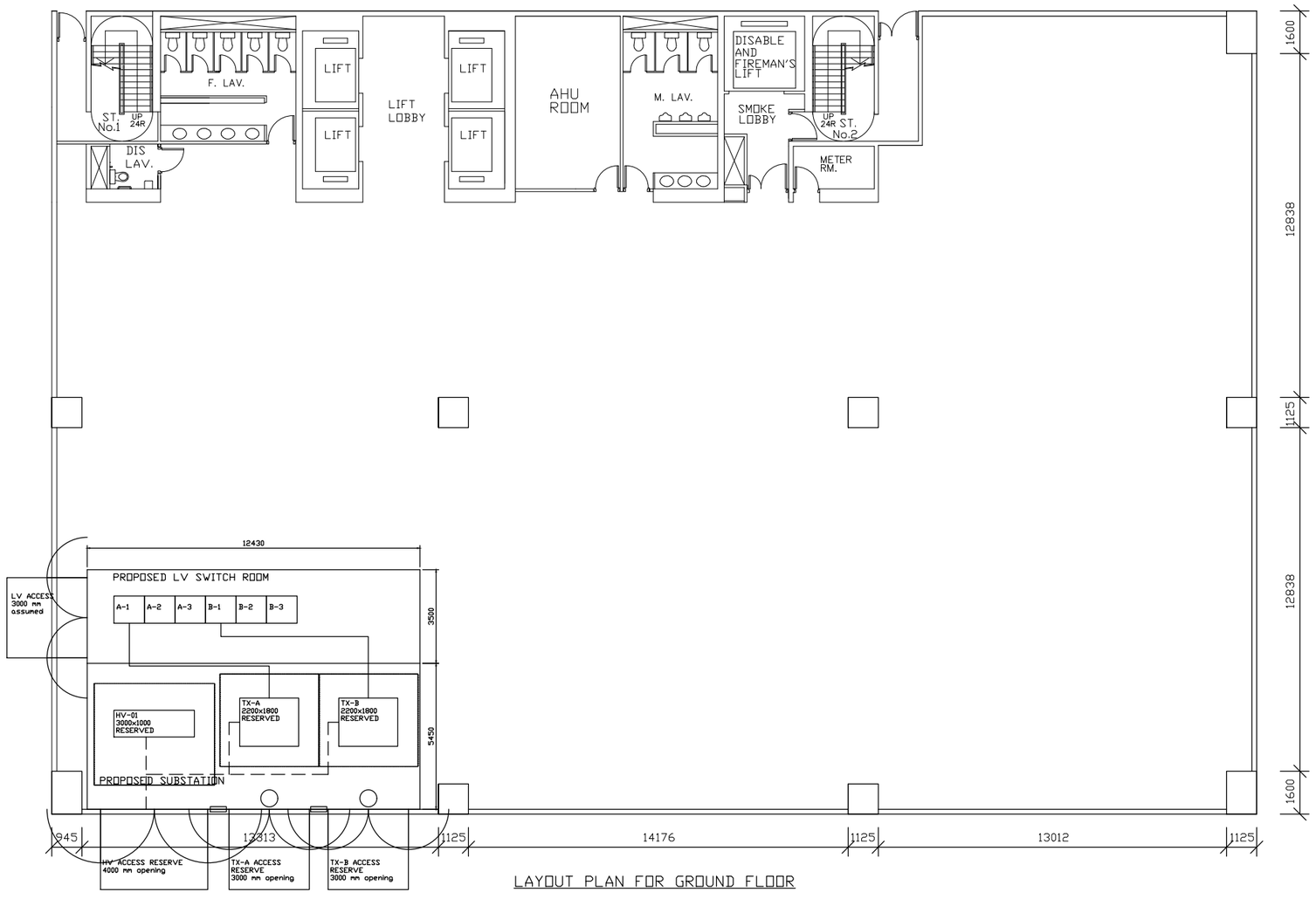
Sources and reproducibility

Cite supplied COP215 Rev 07 and downloaded COP101 v 16 as above. EMSD Wiring CoP2025: Table 6(1), printed 56/PDF 57; 7(1), printed 65–67/PDF 66–68; 9B, printed 79/PDF 80; 11(2), printed 99–100/PDF 100–101; 11(10), printed 105/PDF 106; A5 factors PDF 274–279; A6 tables PDF 282–285. ABB guide 9AKK108468A3912 printed 130 gives the class 3 B-curve energy references. Eaton TDC180 Technical Data 2042 August 2024 p 2 gives conventional-fusing/interruption references, not total energy. URLs and explicit conditions are in the common data file `fault_design_conditions.json`. Tables are generated from current common CSV/JSON data. Reproduce the lighting/power route, demand, incoming and fault scripts before rebuilding this answer. Guarded DWG mutations must not be rerun blindly. Original source documents are retained. Final acceptance still needs normal lighting/power layout PDF review and final DWG save/reopen consistency verification.

Drawing appendix The following sheets show the ground-floor proposal and complete Office 1 DB schematic. Normal lighting/power layouts remain in the editable DWG and will be appended after final plotting review.

TASK 1 | GROUND-FLOOR PLANTROOM LOCATION

PROPOSAL ON SUPPLIED ARCHITECTURAL PLAN | COURSE DESIGN REVIEW | PLAN 1:200 AT A3



LAYOUT PLAN FOR GROUND FLOOR

PROPOSED LOCATION

Lower-left perimeter of supplied ground-floor plan. Substation 12430 x 5450; adjacent LV room 12430 x 3500 assumed. See separate 1:75 room detail.

LOCATION REASONS

Direct external plant access can be reserved subject to site confirmation.

Adjacent LV room keeps transformer-to-incomer interfaces short.

Perimeter plantrooms separate supply equipment from the upper lift/stair/AHU core and remaining accommodation.

LV room has its own proposed west entrance.

LOAD ASSESSMENT

Offices: 1644.800 kVA.

AC: 1000 kW cooling -> 526.039 kW electrical -> 584.488 kVA at assumed PF 0.90.

Public services: 200 kVA aggregate allowance.

Two-zone conservative envelope: 2672.216 kVA.

Propose 2 x 1500 kVA, with equal zone allocation; no full-load N-1.

SITE CONDITIONS TO VERIFY

External apron, road and <=20m unloading route are not given by the source plan. Maintain >=3m clear delivery route.

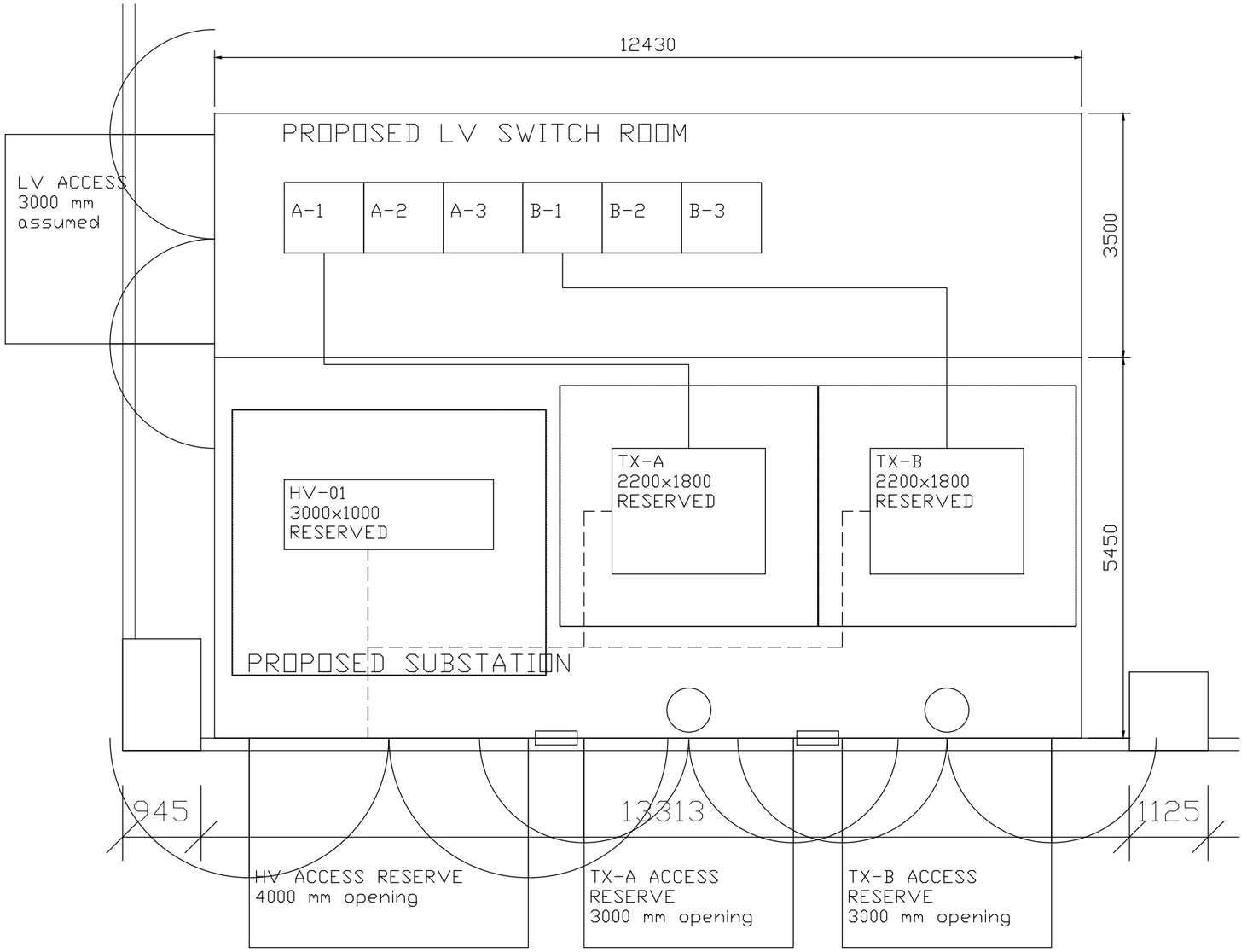
Architectural wall openings, structural/flood/fire provisions, ventilation and supplier equipment approval require coordination.

DRAWING REFERENCES: assignment_working_v02.dwg | TASK1-GROUND-OVERVIEW + TASK1-ROOM-REVIEW

Calculation sources and declared assumptions: Task1_answer_draft.md; data/task1_design_inputs.csv. Original drawings retained.

TASK 1 | PROPOSED SUBSTATION AND MAIN LV ROOM

Office building: 10 typical floors + ground | COURSE DESIGN REVIEW | All dimensions in mm



LINE KEY: dashed = HV trench; solid = LV route; chain = clearance envelope.

LV INTERFACES / HV ROUTES
LV trench: 1000W x 1100D, four MCTs per interface. A: 8.03+2=10.03m; B: 8.31+2=10.31m (<20m). 2m is assumed total termination/vertical allowance.
HV centre-lines: 1000W at panels; 600W branches; final 300W; generally 1400D. Segregate HV/LV. Covered trenches; no beam obstruction.

DETAIL 1:75 AT A3 | NOT A CONSTRUCTION DRAWING

CAPACITY AND EQUIPMENT
2 x 1500 kVA proposed; normal load envelope 2672.216 kVA. No full-load N-1.
HV: 3000 x 1000 reservation.
TX-A/B: 2200 x 1800 each.
These are assumed equipment envelopes, not manufacturer footprints.

CLEARANCES / BUILDING WORKS
HV rear 1000, front 1800, sides 750. TX LV terminals north: 900; other sides 750.
Proposed 2h FRR partitions; fire-seal penetrations. FFL +0.15 vs external +0.00 assumed.
Actual structures, wall thickness and equipment dimensions require coordination.

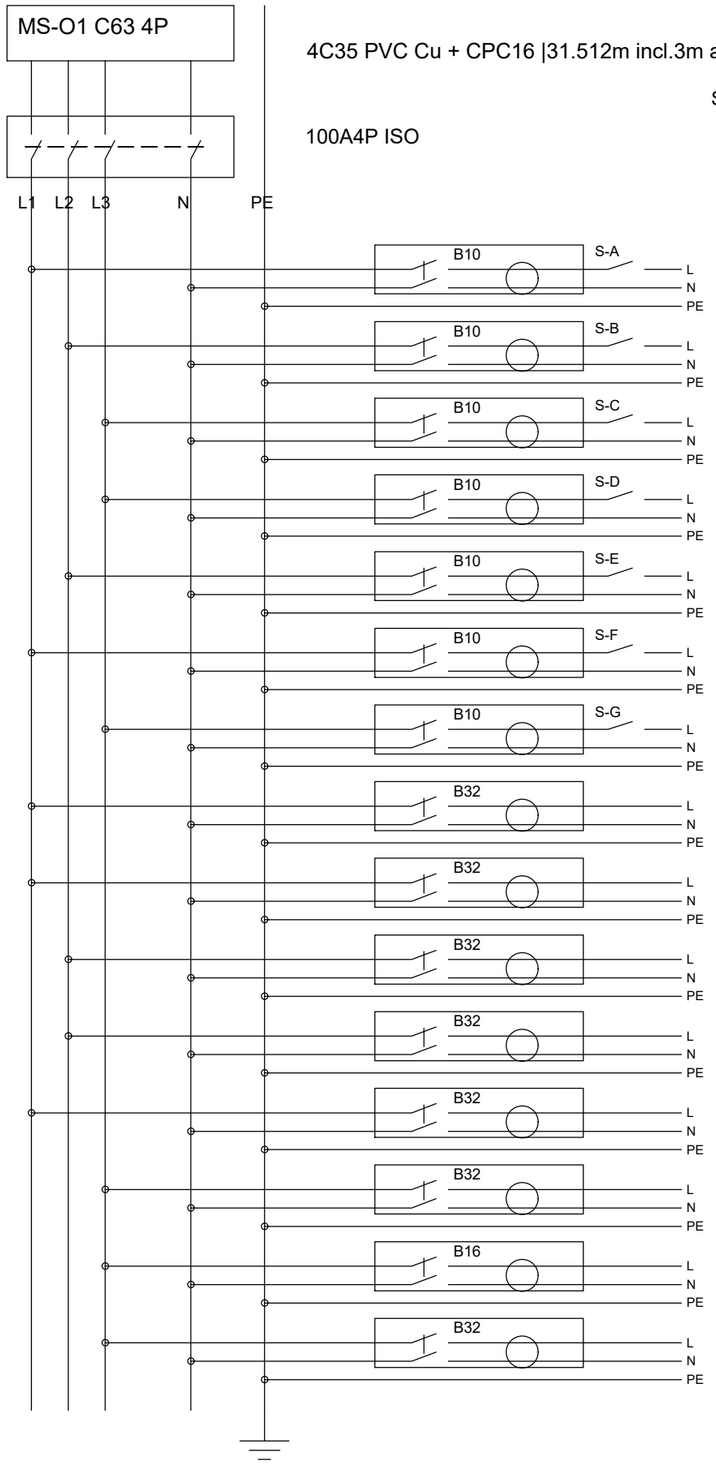
ACCESS AND DOORS
South external access assumed. Reserve 3000-wide delivery route; unloading distance <=20m to verify.
Plant doors: HV 4000W; TX 3000W each; 2800H assumed. 180-degree outward sweep, shown open 90 degrees.
Shared sweeps: use adjacent doors sequentially. TX-A designated wicket personnel access. Independent LV entrance west.

VENTILATION
2 low-level 600W x 2000H inlet louvres: 2.4m2 gross x assumed 50% free area = 1.20m2 >=1.12m2. Certified free area >=46.67% required.
2 high-level 630-dia fan symbols are conditional references. Actual losses, fan duty and noise to confirm; outlet >=2.5m above street, away from people.

Basis: supplied COP215 Rev07; COP101 v16 clauses 4.3, 5.5, 5.6, 6.1-6.7.
See Task1_answer_draft.md and shared data for assumptions/calculations.

TASK 2 | OFFICE 1 DISTRIBUTION BOARD DB-O1

100A TPN + N/PE | 15 OUTGOING CIRCUITS | 7 LIGHTING / 8 POWER | SCHEMATIC - NOT TO SCALE



Source Ze0.20ohm /PFC<=6kA assumed; main/RCBO Icn10kA. See Task3 for conditions and I2t limits.

Landlord emergency/exit systems: independent essential supply assumed; electrically separate from this normal DB.

LC-A [L1] | 18 lamps +S-A /402W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=58.301m | LIGHTING RADIAL

LC-B [L2] | 17 lamps +S-B /392W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=62.184m | LIGHTING RADIAL

LC-C [L3] | 18 lamps +S-C /396W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=44.999m | LIGHTING RADIAL

LC-D [L3] | 11 lamps +S-D /242W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=27.233m | LIGHTING RADIAL

LC-E [L2] | 14 lamps +S-E /392W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=34.464m | LIGHTING RADIAL

LC-F [L1] | 10 lamps +S-F /280W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=32.710m | LIGHTING RADIAL

LC-G [L3] | 4 lamps +S-G /112W | B10 /TYPE A30mA1P+N RCBO
L/N/PE 2.5/2.5/2.5mm2 | L=27.379m | LIGHTING RADIAL

P1 [L1] | FB01, FB02 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 6/6/6mm2 | L=28.620m | A1 RING: L/N/PE BOTH ENDS AT DB

P2 [L1] | FB03, FB04 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 6/6/6mm2 | L=35.520m | A1 RING: L/N/PE BOTH ENDS AT DB

P3 [L2] | FB05, FB06 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 6/6/6mm2 | L=45.220m | A1 RING: L/N/PE BOTH ENDS AT DB

P4 [L2] | FB07, FB08 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 6/6/6mm2 | L=34.220m | A1 RING: L/N/PE BOTH ENDS AT DB

P5 [L1] | SO02 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 10/10/10mm2 | L=15.607m | DEDICATED RADIAL

P6 [L3] | FB09, SO01 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 10/10/10mm2 | L=13.864m | DEDICATED RADIAL

P7 [L3] | FS01 /13A FUSED POINTS | B16 /TYPE A30mA1P+N RCBO
L/N/PE 4/4/4mm2 | L=13.107m | DEDICATED RADIAL

P8 [L3] | SO03 /13A FUSED POINTS | B32 /TYPE A30mA1P+N RCBO
L/N/PE 10/10/10mm2 | L=15.107m | DEDICATED RADIAL

SYMBOLS: box = linked L+N RCBO; circle = residual sensing; dots = junctions; crossings without dots are not connected. PE bypasses all switches.

Each RCBO load neutral is dedicated. Ring L/N/CPC ends return to the same circuit terminals; no mixed neutrals. All local power spurs:13A fuse +4mm2 L/N/CPC,0.5m assumed.

S-A/B/C: separate enclosures for different phases. All manual controls20AX/LED>=600W minimum; local switching and RCBO inrush performance must meet adopted design specification.